

Walking a Private Road: Direction-Guided Perturbation for Trajectory Data Under Local Differential Privacy

Xiaojie Cao¹, Yinbin Miao¹, *Senior Member, IEEE*, Xiaojian Zhang², Zhiquan Liu³, *Senior Member, IEEE*, Jun Feng⁴, *Member, IEEE*, Xinghua Li⁵, and Jianfeng Ma⁶, *Member, IEEE*

I. INTRODUCTION

Abstract—Trajectory data is a crucial resource for analyzing individual behavioral patterns and spatial dynamics. However, directly collecting trajectory data easily lead to privacy leakage issues. Therefore, the mainstream trajectory collection methods use Local Differential Privacy (LDP) to protect trajectory privacy, but they are still suffering from low data utility due to directional discretization and excessive allocation of privacy budget. To address these problems, we propose a basic Direction-Guided Perturbation (DGP) scheme for trajectory, while satisfying ϵ -LDP. Specifically, we utilize the inherent information of trajectory to perturb each trajectory point sequentially and restrict the perturbation region through the direction between adjacent points. To enhance the trajectory utility, we further extend DGP by introducing a radius to constraint the perturbation region, namely R-DGP. By calculating the maximum distance of each trajectory, R-DGP precisely restricts the perturbation range of each trajectory point to enhance response accuracy for data analysis tasks. Finally, we conduct extensive experiments on real-world and synthetic datasets to compare our proposed schemes with state-of-the-art methods. The results show that our schemes reduce the mean normalized error by an average of 20% on four datasets under the same privacy requirement.

Index Terms—Local differential privacy, privacy-preserving, trajectory collection.

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Xiaojie Cao and Yinbin Miao are with the School of Cyber Engineering, Xidian University, Xi'an 710071, China (e-mail: 24151111248@stu.xidian.edu.cn; ybmiao@xidian.edu.cn).

Xiaojian Zhang is with the School of Computer and Information Engineering, Henan University of Economics and Law, Zhengzhou 450046, China (e-mail: xjzhang82@alu.ruc.edu.cn).

Zhiquan Liu is with the College of Cyber Security, Jinan University, Guangzhou 510632, China (e-mail: zqliu@vip.qq.com).

Jun Feng is with the School of Cyber Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: junfeng@hust.edu.cn).

Xinghua Li and Jianfeng Ma are with the State Key Laboratory of Integrated Service Networks, School of Cyber Engineering, Xidian University, Xi'an 710071, China (e-mail: xhli1@mail.xidian.edu.cn; jfma@mail.xidian.edu.cn).

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WITH the widespread application of Location-Based Services (LBS), the trajectory has become an important asset for intelligent transportation, urban planning and personalized services [1], [2]. Since the trajectory inherently contains sensitive information, the direct trajectory collection and analysis may reveal workplace, travel patterns, personal preferences and even identity. Although the anonymization and aggregation techniques to protect trajectory privacy [3], the trajectory remains vulnerable to privacy attacks, such as membership inference [4], [5] and trajectory reconstruction [6]. To address the risk of privacy leakage caused by untrustworthy third-party [7], Local Differential Privacy (LDP) [8] provides a strong privacy protection that is independent of the adversary's background knowledge. Therefore, LDP-based trajectory collecting methods have been extensively explored among academic and industrial fields, which allow users locally to perform privacy processing on data before uploading.

However, the inherent trajectory pattern makes it challenging to achieve rigorous privacy guarantee. To improve trajectory utility, most trajectory collection methods cannot provide strict privacy guarantee of ϵ -LDP, such as geo-indistinguishability [9]. To satisfy ϵ -LDP while guaranteeing high data utility, NGRAM [10] uses the exponential mechanism to directly perturb individual trajectory and utilizes external geographic knowledge and overlapped n-grams to obtain accurate perturbation, but this external knowledge is not always be accessible and can become outdated easily. Outdated or inaccurate geographical knowledge as prior information can lead to severe structural deviations, thereby significantly reducing the trajectory utility. To overcome this limitation, the state-of-the-art mechanism ATP [11] eliminates the reliance on external knowledge and improves the trajectory utility while satisfying ϵ -LDP. Specifically, ATP leverages the correlation between trajectory points, which restricts the perturbation region by replicating trajectory, alternately sampling pivot points, and discretizing the direction of pivot points. Nevertheless, ATP still has certain limitations:

- *Poor data utility*: Dividing the direction of a trajectory point into multiple small discrete radian intervals reduces the representation accuracy of trajectory point direction. Additionally, when ATP perturbs the discretized true direction, it may be perturbed into a direction interval that is

completely opposite to the original direction, which further results in poor trajectory utility.

- *Privacy budget dilution:* To ensure that each point can obtain directional information between adjacent points, ATP replicates each trajectory into two copies which perform exactly the same process. As a result, the two replicas share overall privacy budget, which leads to a smaller privacy budget allocated to each task (perturbing pivot and non-pivot points, perturbing pivot point directions and restricting trajectory region). The smaller the privacy budget is, the larger the noise is. When a trajectory is too long, this may lead to severe perturbation or inaccurate region limitations.

The limitations of existing solutions motivate us to design a new trajectory perturbation scheme. We observe that a user's trajectory is a sequence of spatio-temporal location points, and its movement between consecutive trajectory points requires to be determined in a general direction. Each trajectory point is mainly influenced by its previous point. When a user reaches the current point, the movement direction is determined by the previous point, and the next point is determined by the current point. Therefore, we propose a Direction-Guided Perturbation (DGP) scheme for trajectory collection, which aims to achieve a better balance between privacy protection and data utility. In the process of designing methods, we face two main challenges. **The first challenge is how to preserve the direction continuity between trajectory points as much as possible.** The spatial structure of a trajectory relies heavily on the direction changes between adjacent points. Inappropriate perturbation or direction handling can easily cause the perturbed result to deviate too much from the original direction, leading to inaccurate restriction on the perturbed region. **The second challenge comes from the impact of privacy budget allocation on data utility.** Under user-level LDP, user data must be perturbed locally to protect all privacy information. Therefore, the privacy budget needs to be allocated to each point as well as its additional perturbations, and a low budget produces heavier perturbations when there are too many tasks in a trajectory.

To address the above challenges, unlike NGRAM [10] that discretizes geographical locations and ATP that discretizes direction angles, our method DGP operates directly in the continuous direction space, and introduces a sequence perturbation strategy to maintain the consistency of trajectory direction and overall coherence. Specifically, we first apply perturbations according to the motion direction of adjacent trajectory points. The current point uses the direction of the previous perturbed trajectory point as a reference to determine the perturbation region. This method maximally preserves the directional information of original trajectory and avoids perturbing the direction to be completely opposite to the original motion direction.

Based on the perturbation region determined by direction, trajectory points may still be perturbed within a larger spatial range. Considering that the user's position changes usually have continuity in actual trajectory movement, there always exists an observable maximum movement distance threshold between adjacent points. This observation inspires us to introduce a radius constraint method based on DGP (R-DGP), which restricts each

point within this threshold range. R-DGP ensures that trajectory points are always within a reasonable spatial region, making perturbed trajectory points spatially closer to their original locations. In addition, we conduct extensive experiments using various datasets by comparing our schemes with state-of-the-art solutions. In summary, the key contributions of our work are as follows:

- We construct a new trajectory collection process that utilizes the inherent motion pattern of trajectory to capture the direction of trajectory points to guide perturbation process, which significantly reduces noise while preserving the spatial features of trajectory.
- We propose a new perturbation scheme DGP for trajectory data under LDP, which collects private trajectory data while satisfying ϵ -LDP.
- To enhance the utility, we further enhance DGP by introducing a radius constraint on each point's perturbation region, leading to our optimization method R-DGP.
- We perform theoretical analysis on the perturbation errors and demonstrate that our schemes outperform existing methods through extensive experiments on both real and synthetic datasets. Under the same privacy budget, our schemes reduce the mean normalized error by an average of 20% and the preservation range query exceeds 40% on the CHI, CLE, NYC and CPS datasets.

The rest of this paper is organized as follows. In Section II, we review the related work. In Sections III and IV, we introduce the preliminaries and problem formulation, respectively. In Sections V and VI, we introduce the proposed schemes in detail. In Section VII, we describe the experimental results. Finally, we conclude the paper in Section VIII.

II. RELATED WORK

The collection and analysis of trajectory data based on differential privacy has received widespread attention. Early research primarily focus on frequency estimation, histogram estimation and statistical data analysis [12], [13], [14], [15], [16], [17], [18], [19]. Differential privacy has already been applied in real world such as Google Chrome [20] and Apple's operating system [21]. The above studies and practices provide a solid foundation for the in-depth application trajectory. Currently, the DP or LDP-based trajectory data solutions focusing on trajectory publishing, synthesis and collection have been extensively explored.

Trajectory publishing based-Centralized DP: Many studies focus on trajectory publishing in centralized setting [6], [9], [22], [23], [24], [25], [26], [27]. For example, Jiang et al. [23] introduced the exponential mechanism to sample a suitable direction and distance at each position to ensure the trajectory smoothness. Geo-Indistinguishability [9] is a formal privacy concept for location-based systems that protect the exact locations of users, while allowing approximate information. GL [25] leverages both local and global frequency perturbation mechanisms to enhance the trajectory privacy. To capture more precise information, DPT [27] uses multiple unified grids with different granularities to create different private prefix trees to model transitions and generates data by random walks. AdaTrace [6]

divides geographic locations into two grid layers. The first layer constructs Markov model by leveraging key features such as starting/terminal states, trajectory distribution and length, then the second layer dynamically samples data from this model during the generation phase. Building on AdaTrace and DPT, PrivTrace [26] based on Laplace mechanism [28] adaptively selects first-order and second-order Markov models to balance accuracy and noise, respectively.

Despite the development of various centralized DP algorithms for trajectory publication, these methods rely on a trustworthy aggregator to collect raw trajectories. In contrast, LDP [8] allows users to locally perturb data before uploading it, thereby eliminating the reliance on a trusted data collector and reducing the risk of sensitive information leakage. The research on trajectory data under LDP can be mainly divided into two categories: trajectory synthesis and trajectory collection. Trajectory synthesis aims to extract information from the real statistical distribution, thereby generating a new dataset. On the other hand, trajectory collection focuses on preserving user's continuous movement patterns by directly perturbing the original trajectory points.

Trajectory synthesis based-Local DP: Trajectory synthesis aims to extract the movement patterns of each user under LDP. Therefore, several studies [29], [30], [31] has been dedicated to exploring this field. For example, LDPTTrace [29] uses first-order Markov model to synthesize trajectories by collecting user's trajectory length, transition states and starting/ending points. However, LDPTTrace also faces significant noise issues. In contrast, PLTS [30] further incorporates semantic constraints and uses k -RR [32] to perturb overlapping sub-traces. Although PLTS improves the data accuracy, it incurs high computational cost. Both LDPTTrace and PLTS are compositions based on static data, RetraSyn [31] utilizes OUE to collect transition states and dynamic migration updates for real-time synthetic trajectory generation.

Trajectory collection based-Local DP: In the scenario of trajectory collection, most methods add noise to individual trajectory point. Wang et al. [33] proposed a Staircase Random Response (SRR) based on location-based services, which assigns different perturbation probabilities to locations based on their proximities to true locations. NGRAM [10] is a classic method, which utilizes overlapping n-grams and external knowledge (such as business information and Points of Interest (POIs) categories) to perturb trajectory subsequences. Although utilizing external knowledge can enhance the data practicality, it is difficult to obtain and update. Moreover, in order to achieve precise perturbed trajectories, NGRAM uses linear programming solver to deal with external knowledge, which is time-consuming process. ATP [11], the state-of-the-art trajectory collection method, eliminates the reliance on external knowledge and utilizes bidirectional information between adjacent points to restrict the perturbation region. However, since this method replicates trajectory copies, the privacy budget allocated to multiple tasks in long-trajectory scenarios will be too small, thereby introducing excessive noise.

We compare the scheme proposed in this paper with previous trajectory perturbation schemes, as shown in Table I.

TABLE I
COMPARISON OF OUR PROPOSAL WITH PREVIOUS WORKS

Schemes	External knowledge	Trajectory Replication	Spatial/directional modeling	Privacy Level
LDPTTrace [29]	○	○	Discrete space	ϵ -LDP
PLTS[30]	○	○	—	ϵ -SLDP
NGRAM [10]	●	○	Discrete space	ϵ -LDP
ATP [11]	○	●	Discrete direction	ϵ -LDP
Ours	○	○	Continuous direction	ϵ -LDP

Note: ● represents reliance on external knowledge and trajectory replication, ○ represents non-reliance on external knowledge and the use of a single trajectory.

III. PRELIMINARIES

In this section, we briefly introduce preliminaries on local differential privacy and perturbation mechanisms.

A. Local Differential Privacy

LDP assumes that the data collector is untrustworthy and allows users to locally perturb their own data before uploading them. The data collector then aggregates these reported values in response to queries while preserving user privacy.

Definition 1 (Local Differential Privacy [8]): Given a privacy budget $\epsilon > 0$, a randomized mechanism $\mathcal{M} : \mathcal{X} \rightarrow \mathcal{Y}$ satisfies ϵ -local differential privacy defined by (1) if and only if for any two inputs $x, x' \in \mathcal{X}$ and output $y \in \mathcal{Y}$:

$$\Pr[\mathcal{M}(x) = y] \leq e^\epsilon \times \Pr[\mathcal{M}(x') = y], \quad (1)$$

where the privacy budget ϵ quantifies the level of privacy protection. Thus, a smaller ϵ implies a stronger privacy guarantee, as it is harder for the adversary to distinguish between x and x' . The following properties are crucial for the design and theoretical analysis of our proposed method.

Theorem 1 (Sequential composition [34]): Given a set of n randomized mechanisms $\{\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_n\}$, if each mechanism $\mathcal{M}_i$ ($1 \leq i \leq n$) satisfies ϵ_i -LDP, then the sequential composition of these mechanisms satisfies ϵ -LDP, where $\epsilon = \sum_{i=1}^n \epsilon_i$.

Theorem 2 (Post-Processing [34]): Let $\mathcal{M}$ be a randomized algorithm that satisfies ϵ -LDP, and g be an arbitrary randomized mapping. Then, $g(\mathcal{M}(\cdot))$ satisfies ϵ -LDP.

The sequential composition guarantees the overall privacy of the random algorithm sequences. Furthermore, the post-processing property ensures that applying any arbitrary computation to the output of an LDP algorithm does not incur additional privacy loss.

B. Perturbation Mechanisms

Randomized Response (RR) [35] is a mainstream perturbation mechanism in LDP. It leverages the uncertainty in responses to sensitive questions to protect original data privacy. The k -RR [32] mechanism extends RR mechanism to support cases where the domain size exceeds two.

Definition 2 (k -RR [32]): Given privacy budget ϵ , k -RR mechanism $\mathcal{M}$ satisfies ϵ -LDP, if for any input $x \in \mathcal{D}$, the

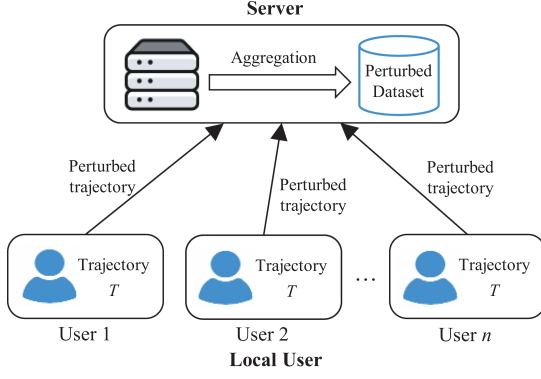


Fig. 1. System model of our schemes.

probability of any output y is defined by (2):

$$\Pr[\mathcal{M}(x) = y] = \begin{cases} p = \frac{e^\epsilon}{e^\epsilon + d - 1}, & \text{if } y = x, \\ q = \frac{1}{e^\epsilon + d - 1}, & \text{if } y \neq x, \end{cases} \quad (2)$$

where p is the output probability of true value, and $q = \frac{1-p}{d}$ is the output probability of any other value.

The k -RR is mainly used for categorical value perturbation, while numerical perturbation methods such as Laplace mechanism [28] and Square-Wave mechanism (SW) [16] are typically applied in LDP settings. The Laplace mechanism introduces noise sampled from a Laplace distribution, but it can produce unbounded noise, which may degrade data utility. In contrast, the SW mechanism is designed to perturb numerical value while ensuring that the noise remains within a bounded range, thereby improving privacy protection and data utility.

Definition 3 (Square-Wave Mechanism [16]): The square-wave mechanism $\mathcal{M}$ satisfies ϵ -LDP, for any input $x \in [0, 1]$ and output $y \in [-b, b + 1]$, the output probability of y is:

$$\Pr[\mathcal{M}(x) = y] = \begin{cases} p = \frac{e^\epsilon}{2be^\epsilon + 1}, & \text{if } |x - y| \leq b, \\ q = \frac{1}{2be^\epsilon + 1}, & \text{otherwise,} \end{cases} \quad (3)$$

where $b = \frac{\epsilon e^\epsilon - e^\epsilon + 1}{2e^\epsilon(e^\epsilon - 1 - \epsilon)}$.

The SW mechanism takes an input value in $[0, 1]$ and outputs a value bounded within $[-b, b + 1]$, effectively limiting the impact of individual data points on query results.

IV. PROBLEM FORMALIZATION

In this section, we present the system model, problem definition and design goals.

A. System Model

The system model of our proposed schemes consists of local user and server, as shown in Fig. 1. The role of each entity is described as follows:

- **Local User:** Each user has a private trajectory, which is perturbed using a mechanism that satisfies ϵ -LDP to protect individual privacy. All perturbed trajectories are then sent to the server (i.e., data collector).

- **Server:** On the server side, upon receiving perturbed trajectories from users, various data analysis tasks are performed based on them, such as preservation range queries and hotspot detection.

B. Problem Definition

We consider in the trajectory space $\mathcal{P}$, each user u has a trajectory $T = \{p_1, p_2, \dots, p_t\}$, where each trajectory point $p_i = (lon_i, lat_i)$ represents a location point with longitude and latitude information. We assume that the trajectory space $\mathcal{P}$ is public and accessible to both data collectors and all users [11]. This is a practical assumption in research on location privacy protection [9], [11], [36]. Under the LDP framework, all users need to locally protect their private trajectories and then send all perturbed trajectories to the server. Therefore, the problem we focus on is how to generate a perturbed trajectory $\hat{T}$ that has a similar spatial pattern to the original trajectory T while satisfying ϵ -LDP. To this end, we design a new trajectory collection scheme to determine the perturbation region D_i for each trajectory point p_i , and perturb the point p_i within the region D_i to generate the perturbed point $\hat{p}_i$. Each user locally uploads his/her perturbed trajectory $\hat{T} = \{\hat{p}_1, \hat{p}_2, \dots, \hat{p}_t\}$ to the server. The server collects all noisy trajectories to support various location-based data analysis tasks. A representative application is hotspot detection, which aims to identify geographic regions with high user visit frequency.

As one of the state-of-the-art trajectory perturbation methods, ATP replicates trajectory T as $T' = T'' = T$. Subsequently, T' and T'' alternately select a set of pivot points that can obtain direction information. Specifically, when perturbing a trajectory point p_i , ATP utilizes its neighboring perturbed points $\hat{p}_{i-1}$ and $\hat{p}_{i+1}$ to determine the perturbation region. The directions of $\hat{p}_{i-1}$ and $\hat{p}_{i+1}$ are discretized into a set of radian intervals $\mathcal{D} = \{0, 1, 2, \dots, |\mathcal{D}| - 1\}$ under a given granularity. Therefore, the perturbation region of p_i is defined as the intersection of directions constraint by this point to $\hat{p}_{i-1}$ and $\hat{p}_{i+1}$. However, the discretization process may lead to significant directional deviations, such as substantial drift or even completely opposite orientations.

C. Design Goals

We aim to design trajectory collection method that has high utility and strong privacy guarantee under LDP. Based on this, our methods should achieve following goals:

- **Trajectory privacy protection:** Our primary goal is to provide *user-level* ϵ -LDP for each user by perturbing every point in his/her trajectory. In other words, our methods ensure that an adversary cannot infer any true location information with high confidence from the released perturbed trajectory.
- **High data utility:** Under the rigorous privacy protection of LDP, our proposed methods should control the amount of noise used for perturbation so as to preserve the utility of perturbed trajectory as much as possible, which is important for many location-based applications.

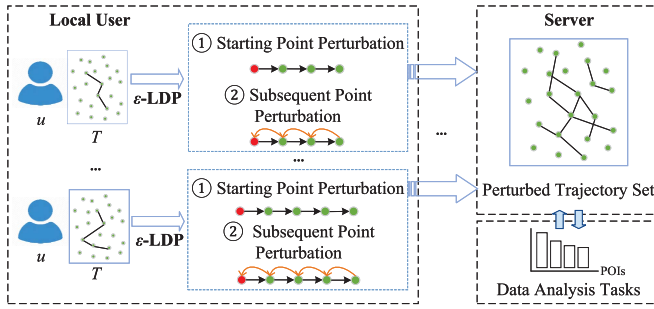


Fig. 2. The overall flow of our proposed schemes.

V. OUR PROPOSED SCHEME

In this section, we first introduce the overall framework of DGP (Direction-Guided Perturbation). Then, in Section B, we present the implementation process of DGP scheme. In Section C, we further propose an optimization strategy based on DGP, called R-DGP, which can more effectively preserve the motion patterns of trajectories and enhance the accuracy of perturbation regions.

A. Technical Overview of Our Schemes

Our goal is to maximize the trajectory utility while ensuring privacy. To achieve this, we introduce a novel perturbation framework based on directional continuity and spatial correlation, which leverages the inherent structural dependencies in trajectory data. The design process of our schemes is presented in Fig. 2. As opposed to the traditional approach which duplicates trajectory copies, we instead directly perturb the trajectory itself to avoid excessive budget allocation. Considering that the direction of subsequent trajectory point highly relies on the previous point, reasonably reducing the perturbation region can improve accuracy. Therefore, we design two strategies: one is a direction-constrained perturbation regions, and the other is a radius-constrained strategy combined with direction optimization.

- **Direction-based:** Existing methods typically discretize the direction between adjacent trajectory points into multiple candidate directions and determine the perturbation region for each point by perturbing the true direction. However, the discretized perturbation directions may deviate from or even be completely opposite to the true direction. To solve this problem and improve the structural fidelity of the perturbed trajectory, we introduce a sequential dynamic constraint mechanism within the continuous direction space. Specifically, we determine the perturbation region for the next trajectory point based on the continuous direction of previous perturbed point. First, the starting point p_1 is perturbed directly in the trajectory space $\mathcal{P}$. For each subsequent trajectory point $p_i (i \geq 2)$, we determine the perturbation region D_i based on the angle $\theta_{i-1,i}$ formed with its previous perturbed point $\hat{p}_{i-1}$. Within the corresponding perturbation region D_i , the perturbed point $\hat{p}_i$ is generated to achieve the continuous constraint perturbation of the trajectory.

- **Direction-optimized radius constraint:** In order to further improve the accuracy of perturbed trajectory, we employ a radius-based region restriction method on top of the direction constraint. This idea comes from the fact that there is always a maximum distance during trajectory movement, so we set the perturbation value of the maximum movement distance $d_{\max}$ between adjacent points in original trajectory as the radius of trajectory points, thereby narrowing down the perturbation region based on directional constraints. After determining the perturbation region D_i for each trajectory point p_i , we apply the k -RR mechanism to perturb trajectory points.

Finally, the perturbed trajectory $\hat{T} = \{\hat{p}_1, \hat{p}_2, \dots, \hat{p}_t\}$ is sequentially aggregated through all perturbed points. The server uses all perturbed trajectories to perform location-related data analysis tasks. The detailed processes for the above two strategies will be introduced in Sections B and C.

B. Implementation of DGP

In this section, we present our proposed DGP scheme in detail. The scheme mainly describes how to utilize the directional information between adjacent trajectory points to determine the perturbation region. To illustrate the necessity and superiority of DGP, we begin with a straightforward approach as a baseline.

Strawman approach. k -RR: The strawman method employs the simplest processing strategy by independently perturbing each trajectory point within the entire trajectory space using the k -RR mechanism. Given a trajectory $T = \{p_1, p_2, \dots, p_t\}$, where each point $p_i (1 \leq i \leq t)$ is represented by its geographic coordinates $p_i = (lon_i, lat_i)$. Specifically, we use budget ϵ/t to perturb each point $p_i \in T$ within the trajectory space $\mathcal{P}$.

The method perturbs each trajectory point within the entire trajectory space, resulting in a significant spatial deviation between the generated noisy trajectory and the original trajectory. To address this issue, we design a perturbation strategy with stronger structural constraints, aiming to enhance the trajectory utility while ensuring privacy. DGP divides the perturbation of trajectory points into two cases: starting point perturbation and subsequent point perturbation. The following provides a detailed introduction to the specific implementation process.

1) **Starting Point Perturbation:** We perturb trajectory points based on the forward path of trajectory T . The p_1 serves as the starting point of trajectory T and has no predecessor for reference, we directly perform the perturbation within the trajectory $\mathcal{P}$ to achieve privacy protection. The trajectory point p_1 is perturbed by (4).

$$\Pr[k-RR(p_1) = \hat{p}_1] = \begin{cases} p = \frac{e^{\epsilon_1/t}}{e^{\epsilon_1/t} + |D_1| - 1}, & \text{if } \hat{p}_1 = p_1, \\ q = \frac{1}{e^{\epsilon_1/t} + |D_1| - 1}, & \text{if } \hat{p}_1 \neq p_1, \end{cases} \quad (4)$$

where $\hat{p}_1 \in D_1$, and D_1 denotes perturbation region of $\hat{p}_1$. In this case, D_1 is defined as $\mathcal{P}$. To illustrate the perturbation process, we take a trajectory $T = \{p_1, p_2, p_3, p_4\}$ as an example (see Fig. 3(a)). The perturbed result of point $\hat{p}_1$ is shown in Fig. 3(b).

2) **Subsequent Point Perturbation:** For each subsequent trajectory point $p_i (i \geq 2)$, the perturbation region can be determined with reference to the previous point direction. Since the

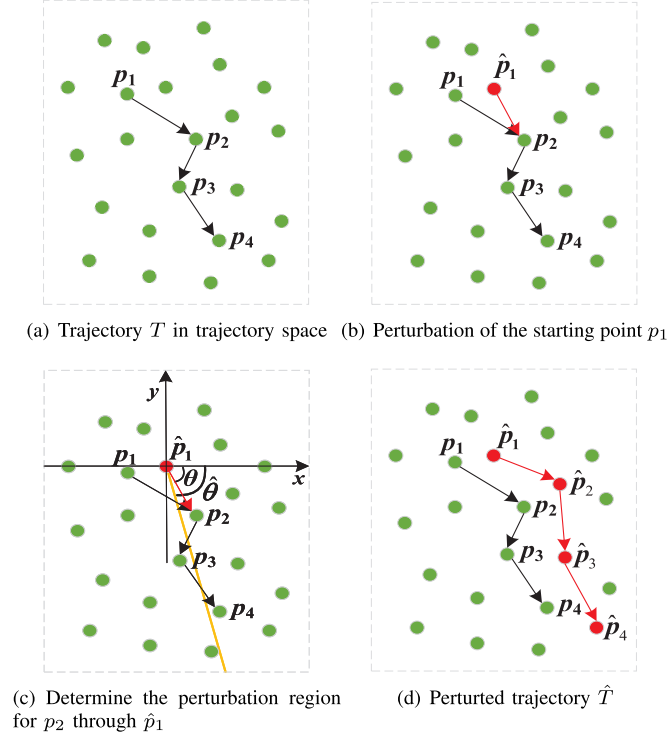


Fig. 3. The perturbation process of trajectory T .

perturbation is applied sequentially to each trajectory point, the perturbation region of each point p_i ($i \geq 2$) is determined by the direction angle $\theta_{i-1,i}$ between perturbed point $\hat{p}_{i-1}$ and p_i , where $\hat{p}_{i-1} = (\text{lon}_{i-1}, \text{lat}_{i-1})$ and $p_i = (\text{lon}_i, \text{lat}_i)$. In order to accurately calculate the directional angle $\theta_{i-1,i}$ between points $\hat{p}_{i-1}$ and p_i , we convert the latitude and longitude of trajectory points into Cartesian coordinates on the x - y plane. We use the Earth's radius $R_E = 6371$ km to scale the geographic coordinates into a planar representation. The process of converting latitude and longitude into Cartesian coordinates is shown in (5).

$$\begin{aligned} x &= R_E \cdot \sin\left((90 - \text{lat}) \cdot \frac{\pi}{180}\right) \cdot \cos\left(\text{lon} \cdot \frac{\pi}{180}\right), \\ y &= R_E \cdot \sin\left((90 - \text{lat}) \cdot \frac{\pi}{180}\right) \cdot \sin\left(\text{lon} \cdot \frac{\pi}{180}\right), \end{aligned} \quad (5)$$

where the term $\sin((90 - \text{lat}) \cdot \frac{\pi}{180})$ converts the geographic latitude into a polar angle measured in radians, which is required for spherical coordinate transformation.

The obtained x and y coordinates provide a planar representation of geographical locations, facilitating spatial analysis and calculations. For example, analyzing the points distribution on a two-dimensional plane and calculating distance between points all rely on the mathematical processing of projected coordinates. Through coordinate transformation, the perturbed point can be represented as $\hat{p}_{i-1} = (\hat{x}_{i-1}, \hat{y}_{i-1})$, and the current point is represented as $p_i = (x_i, y_i)$, where x_i and y_i are the projection coordinates on the two-dimensional plane. The vector from $\hat{p}_{i-1}$ to p_i can be represented as $\mathbf{v}_{i-1,i} = (x_i - \hat{x}_{i-1}, y_i - \hat{y}_{i-1})$. Next, the direction angle $\theta_{i-1,i}$ between vector $\mathbf{v}_{i-1,i}$ and the

Algorithm 1: Get Point Region.

Input: Current point p_i , perturbed point $\hat{p}_{i-1}$, privacy budget ϵ_2
Output: The perturbation region D_i of the p_i

- 1 Calculate Cartesian coordinates of $\hat{p}_{i-1}$ and p_i by (5);
- 2 $\mathbf{v}_{i-1,i} = (x_i - \hat{x}_{i-1}, y_i - \hat{y}_{i-1})$;
- 3 $\theta_{i-1,i} = \arctan2(\frac{y_i - \hat{y}_{i-1}}{x_i - \hat{x}_{i-1}})$;
- 4 $\theta_{i-1,i} = (\theta_{i-1,i} + 2\pi) \pmod{2\pi} \in [0, 2\pi]$;
- 5 $\rho_{i-1,i} \leftarrow \frac{\theta_{i-1,i}}{2\pi} \in [0, 1]$;
- 6 Perturb $\rho_{i-1,i}$ to $\hat{\rho}_{i-1,i}$ by (3) with budget $\epsilon_2/(t-1)$;
- 7 $\hat{\theta}_{i-1,i} = 2\pi \cdot (\frac{\hat{\rho}_{i-1,i} + b}{2b+1})$;
- 8 return D_i .

Algorithm 2: DGP Scheme.

Input: Trajectory $T = \{p_1, p_2, \dots, p_t\}$, trajectory space $\mathcal{P}$, privacy budget ϵ
Output: Perturbed trajectory $\hat{T}$

- 1 Initialize $\hat{T} \leftarrow \emptyset$; ϵ_1, ϵ_2 ;
- 2 **for** $1 \leq i \leq t$ **do**
- 3 **if** $i = 1$ **then**
- 4 $\hat{p}_1 \leftarrow k\text{-RR}(p_1, \mathcal{P}, \epsilon_1/t)$;
- 5 $\hat{T} = \hat{T} \cup \hat{p}_1$;
- 6 **else**
- 7 $D_i \leftarrow \text{GetPointRegion}(p_i, \hat{p}_{i-1}, \epsilon_2)$;
- 8 $\hat{p}_i \leftarrow k\text{-RR}(p_i, D_i, \epsilon_1/t)$;
- 9 $\hat{T} = \hat{T} \cup \hat{p}_i$;
- 10 **end**
- 11 **end**
- 12 return $\hat{T}$.

positive x -axis is calculated using the $\arctan2$ function [23]. The specific calculation formula can be measured by (6).

$$\theta_{i-1,i} = \arctan2\left(\frac{y_i - \hat{y}_{i-1}}{x_i - \hat{x}_{i-1}}\right). \quad (6)$$

In the Cartesian coordinate system, the positive direction of the x -axis is taken as the default angle origin, and the counter-clockwise direction is defined as positive. Since the return value obtained from (6) is within the interval $[-\pi, \pi]$, in order to maintain consistency in the angle range, we need to normalize the result and convert it to the interval $[0, 2\pi]$ for subsequent direction calculations and comparisons. To guarantee mathematical rigor and avoid implementation errors, we define this mapping step as $\theta_{i-1,i} = (\theta_{i-1,i} + 2\pi) \pmod{2\pi}$. Next, we use SW to perturb $\theta_{i-1,i}$ and protect its privacy. According to Definition 3, the angle $\theta_{i-1,i}$ is normalized to $\rho_{i-1,i} = \theta_{i-1,i}/2\pi \in [0, 1]$. Subsequently, $\rho_{i-1,i}$ is perturbed to $\hat{\rho}_{i-1,i}$ through (3). Finally, the perturbed value $\hat{\rho}_{i-1,i}$ is mapped back to the original angle interval $[0, 2\pi]$ using (7) to ensure the validity of perturbed angle.

$$\hat{\theta}_{i-1,i} = 2\pi \cdot \left(\frac{\hat{\rho}_{i-1,i} + b}{2b+1}\right). \quad (7)$$

According to the perturbed angle $\hat{\theta}_{i-1,i}$ measured from the positive x -axis, we construct the corresponding directional angle

region D_i for each trajectory point $p_i (i \geq 2)$. To ensure strict privacy guarantees, this angle is derived relying solely on the previous noisy point $\hat{p}_{i-1}$, without any involvement of the original trajectory point p_{i-1} . This process essentially belongs to a post-processing operation and thus does not consume additional privacy budget. Ultimately, this region is derived from the local trajectory movement direction calculated by the user, and it serves to constrain the possible perturbation domain of the trajectory point.

The specific determination process of directional angle region D_i is shown in Algorithm 1. Essentially, the spatial region defined by the perturbed direction angle $\theta_{i-1,i}$ constitutes the perturbation domain D_i of point $p_i (i \geq 2)$. For instance, as shown in Fig. 3(c), we take the previously perturbed point $\hat{p}_1$ as the origin and calculate the true angle $\theta_{1,2}$ of the vector from $\hat{p}_1$ to p_2 relative to the positive x -axis (Lines 1-4). Then, we obtain the perturbed angle $\hat{\theta}_{1,2}$ (Lines 5-7). The directional angle region D_2 of point p_2 extends from the positive x -axis to the orange line (Line 8). Subsequently, we use k -RR to perturb each trajectory point $p_i (i \geq 2)$ within its corresponding domain D_i to obtain $\hat{p}_i \in D_i$. Therefore, the probability of $p_i (i \geq 2)$ is perturbed to $\hat{p}_i$ is given by (8).

$$\Pr[k - RR(p_i) = \hat{p}_i] = \begin{cases} p = \frac{e^{\epsilon_1/t}}{e^{\epsilon_1/t} + |D_i| - 1}, & \text{if } p_i = \hat{p}_i, \\ q = \frac{1}{e^{\epsilon_1/t} + |D_i| - 1}, & \text{if } p_i \neq \hat{p}_i. \end{cases} \quad (8)$$

The above procedure is executed for each point in trajectory T to construct its corresponding perturbation region, ultimately generating the final perturbed trajectory $\hat{T}$. The overall process of DGP scheme is formally described in Algorithm 2. As illustrated in Fig. 3(d), an example of a perturbed trajectory is given by $\hat{T} = \{\hat{p}_1, \hat{p}_2, \hat{p}_3, \hat{p}_4\}$. Upon receiving the noisy trajectory submitted by users, the server cannot effectively infer the original true angle $\theta_{i-1,i}$, as it has already been perturbed by SW mechanism. For any arbitrary trajectory point p_i , its candidate perturbation region D_i depends on this noisy angle $\hat{\theta}_{i-1,i}$. Once the candidate region D_i is established, the k -RR mechanism is employed to perturb the true trajectory point p_i within this confined domain. Specifically, for any given true location p_i , the conditional probability ratio of the mechanism outputting the true location ($\hat{p}_i = p_i$) versus outputting any other false location within region D_i ($\hat{p}_i \neq p_i$) satisfies $\frac{\Pr[\hat{p}_i = p_i | p_i]}{\Pr[\hat{p}_i \neq p_i | p_i]} \leq e^{\epsilon_1/t}$. Consequently, an adversary cannot confidently infer the true location based solely on the observed noisy results.

Remark: DGP scheme determines the perturbation region of each trajectory point by the direction angles between adjacent points. Compared to the strawman method, DGP significantly reduces the perturbation range and effectively improves the accuracy of noisy trajectories. However, although the directional constraint narrows down the perturbation region, the inherently large spatial scale of the original trajectory makes it difficult for a single directional constraint to sufficiently compress the overall spatial space. Consequently, this limitation can lead to substantial spatial deviations of the perturbed points, disrupting the continuity of the trajectory and ultimately degrading the utility of the noisy data in downstream tasks.

C. Radius Constrained DGP

To address the limitation of DGP, we propose a Radius Constrained Direction-Guided Perturbation (R-DGP) scheme. Unlike DGP, R-DGP restricts the movement of trajectory points within predefined spatial boundaries, ensuring that all points remain within a controllable and reasonable region. By imposing these constraints, R-DGP retains the structural characteristics of trajectory more effectively while improving data utility.

We observe that the movement patterns of objects exhibit significant local continuity by analyzing the characteristics of motion trajectories. Consequently, the distance between any two consecutive points in the trajectory is naturally bounded. To capture this property, we define $d_{\max}$ as the maximum distance between adjacent points in a given trajectory T . It should be emphasized that for any trajectory, there exists only one unique global maximum distance. By utilizing $d_{\max}$ as a radius constraint, we can strictly limit the possible perturbation range of each trajectory point. This constraint effectively minimizes the spatial space, thereby significantly improving the utility of the perturbed data. Specifically, $d_{\max}$ is formally defined as:

$$d_{\max} = \max_{p_i, p_{i+1} \in T} \text{dist}(p_i, p_{i+1}), \quad (9)$$

where p_i, p_{i+1} represent adjacent trajectory points in trajectory T , and $\text{dist}(\cdot)$ is the Haversine distance. The Haversine distance calculates the great-circle distance, which is the shortest distance along the surface of the Earth. Note that throughout this paper, all distance calculations employ the Haversine distance and are uniformly measured in kilometers (km).

To ensure privacy, the maximum distance $d_{\max}$ must be perturbed. Since the distance is a continuous numerical value, a straightforward approach is to apply Laplace mechanism to $d_{\max}$. This requires calculating the global sensitivity, which corresponds to the maximum possible distance between any two points in the entire location space $\mathcal{P}$. The sensitivity Δd can be formally defined as:

$$\Delta d = \max_{p_i, p_{i+1} \in \mathcal{P}} \text{dist}(p_i, p_{i+1}). \quad (10)$$

However, the Laplace mechanism suffers from unbounded noise. A large sensitivity Δd amplifies this noise, which not only leads to significant utility degradation but can also produce unrealistic negative distances.

We use SW mechanism to obtain the perturbed maximum distance $\hat{d}_{\max}$, and ensure a bounded perturbation range. Then, we use the $\hat{d}_{\max}$ as a radius constraint to bound the perturbation region for each point $p_i (i \geq 2)$. Specifically, we first normalize $d_{\max}$ into the finite domain $[0, 1]$ via $r = d_{\max} / \Delta d$, where Δd serves as the theoretical upper bound of $d_{\max}$. Subsequently, we perturb r to obtain $\hat{r}$ using (3), and then map the perturbed value $\hat{r}$ back to the original distance range via (11):

$$\hat{d}_{\max} = \frac{(\hat{r} + b)\Delta d}{2b + 1}. \quad (11)$$

The overall R-DGP procedure is formally detailed in Algorithm 3. For a given trajectory T , the algorithm first computes the maximum distance $d_{\max}$ and perturbs it with privacy budget ϵ_3 to obtain $\hat{d}_{\max}$. The perturbation of the starting point

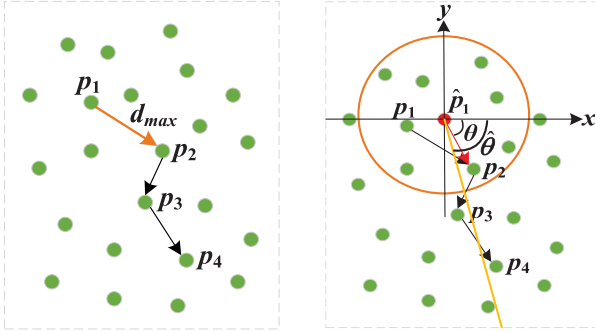
Algorithm 3: R-DGP Scheme.

Input: Trajectory $T = \{p_1, p_2, \dots, p_t\}$, trajectory space $\mathcal{P}$, privacy budget ϵ

Output: Perturbed trajectory $\hat{T}$

```

1 Initialize  $\hat{T} \leftarrow \emptyset$ ;  $\epsilon_1, \epsilon_2, \epsilon_3$ ;
2  $d_{\max} = \max_{p_i, p_{i+1} \in T} \text{dist}(p_i, p_{i+1})$ ;
3 Perturb  $d_{\max}$  to  $\hat{d}_{\max}$  by Eq. 3 with budget  $\epsilon_3$ ;
4 for  $1 \leq i \leq t$  do
5   if  $i = 1$  then
6      $\hat{p}_1 \leftarrow k\text{-RR}(p_1, \mathcal{P}, \epsilon_1/t)$ ;
7      $\hat{T} = \hat{T} \cup \hat{p}_1$ ;
8   else
9     Get the region  $P_r^i$  centered at  $\hat{p}_{i-1}$  with radius
        $\hat{d}_{\max}$ ;
10     $D_i \leftarrow \text{GetPointRegion}(p_i, \hat{p}_{i-1}, \epsilon_2)$ ;
11     $\hat{p}_i \leftarrow k\text{-RR}(p_i, P_r^i \cap D_i, \epsilon_1/t)$ ;
12     $\hat{T} = \hat{T} \cup \hat{p}_i$ ;
13  end
14 end
15 return  $\hat{T}$ .
```



(a) The maximum distance between any neighbors in trajectory T (b) Perturbation region of p_2 under radius constraint

Fig. 4. Illustration of R-DGP scheme.

remains identical to that in DGP (Lines 5-7). Since there is no previous reference point at the starting point, we will not apply the radius constraint to it. For each subsequent point p_i ($i \geq 2$), R-DGP introduces a radius to restrict the perturbation range (Line 9). Specifically, the algorithm uses the previously perturbed point $\hat{p}_{i-1}$ as the center of a circle with $\hat{d}_{\max}$ as the radius, thus forming a circular region P_r^i . The final perturbation region D_i is then defined as the geometric intersection of the circular region P_r^i and the candidate region generated by Algorithm 1 (Lines 10-11). For example, as illustrated in Fig. 4(a), $d_{\max}$ represents the maximum true distance between any two consecutive points in T (indicated by the orange line). Following perturbation, $\hat{d}_{\max}$ serves as the radius of a circular centered at $\hat{p}_1$, as shown in Fig. 4(b). Consequently, the refined perturbation region D_2 for point p_2 is precisely determined by the intersection of the circular constraint and the direction-guided region, which geometrically corresponds to the sector enclosed by the yellow boundary and the positive x -axis.

It should be noted that the maximum distance $d_{\max}$ is computed exactly once before the perturbation process begins. After the perturbation, the obtained noisy value $\hat{d}_{\max}$ will be applied to the global radius constraints of all subsequent trajectory points. Therefore, the radius constraints will neither be recalculated independently nor be dynamically updated for each point p_i . This one-time calculation design not only imposes a unified spatial boundary throughout the trajectory generation process, but also effectively avoids the repeated consumption of the privacy budget and minimizes the computational cost.

We find that utilizing the perturbed radius $\hat{d}_{\max}$ to constrain the region effectively prevents perturbed points from deviating excessively from their true locations. In practical scenarios, the movement of objects (e.g., vehicles and pedestrians) is physically constrained by factors such as speed, direction, and geographical topologies, thereby following specific spatial patterns. The positional shifts between consecutive trajectory points are typically gradual, resulting in relatively small distance variations. By introducing radius constraint, we ensure that the spatial displacement of each perturbed point respects the maximum realistic distance bound of the trajectory. This constraint-based strategy not only preserves the intrinsic motion patterns and local geometric features of the original trajectory but also significantly enhances the overall utility and logical consistency of the perturbed data.

VI. PRIVACY AND UTILITY ANALYSIS

In this section, we formally analyze the privacy and utility of DGP and R-DGP. In Theorems 3 and 4, we establish the privacy guarantee of two schemes. Then, we further conduct a theoretical analysis of the variance introduced by their perturbation mechanisms.

Theorem 3: DGP satisfies ϵ -LDP for each user.

Proof: For any trajectory $T = \{p_1, p_2, \dots, p_t\}$ of length t , Algorithm 1 presents the details of DGP. The overall privacy budget ϵ is divided into two parts in order to protect trajectory points and direction angles: ϵ_1 is allocated for point perturbations, and ϵ_2 is allocated for direction angle perturbations.

For the trajectory point perturbations, we use k -RR with privacy budget ϵ_1/t to independently perturb each point. Each independent perturbation of trajectory points satisfies ϵ_1/t -LDP. The total number of point perturbations is t . For each point p_i ($i \geq 2$), its direction angle $\theta_{i-1,i}$ is formed based on the position of the previously perturbed point $\hat{p}_{i-1,i}$, which serves as the origin. It should be emphasized that the calculation of any direction angle is entirely based on the perturbed point and does not access the original point. Therefore, the operation of constructing the direction based on noisy data belongs to the post-processing step. According to the post-processing property of LDP, this process does not consume any additional privacy budget. Each perturbation for direction angles by SW satisfies $\epsilon_2/(t-1)$ -LDP. The total number of direction angles is $t-1$.

By applying the sequential composition theorem, the overall privacy budget is $\sum_{i=1}^t \epsilon_1/t + \sum_{i=1}^{t-1} \epsilon_2/(t-1) = \epsilon$. Therefore, the entire DGP scheme guarantees ϵ -LDP. $\square$

We analyze the error introduced by the perturbation mechanism in DGP scheme for tasks such as hotspot detection. There are X users who share the same restricted perturbation domain D_i at point p_i . To estimate the frequency of a specific target location p_i , let $c'_X(p_i)$ denote the total observed count of users among X who report their trajectory point as p_i . The estimate count of $c'_X(p_i)$ is denoted as $\hat{c}_X(p_i)$ and can be computed as follows:

$$\hat{c}_X(p_i) = \frac{c'_X(p_i) - X \cdot q}{p - q}, \quad (12)$$

where p represents the probability (i.e., $e^{\epsilon_1/t}/(e^{\epsilon_1/t} + |D_i| - 1)$) that true point is reported, and q represents the other probability (i.e., $1/(e^{\epsilon_1/t} + |D_i| - 1)$).

The expected value of this estimate is $E[\hat{c}_X(p_i)] = c_X(p_i)$, which indicates that the result is unbiased. The variance of the estimated count $\hat{c}_X(p_i)$ can be derived as follows:

$$\text{Var}[\hat{c}_X(p_i)] = X \frac{q(1-q)}{(p-q)^2} = X \frac{|D_i| - 2 + e^{\epsilon_1/t}}{(e^{\epsilon_1/t} - 1)^2}. \quad (13)$$

Next, we analyze the error caused by the perturbation of directional angle $\theta_{i-1,i}$ between $\hat{p}_{i-1}$ and p_i . Since $\rho_{i-1,i}$ is a normalized version of $\theta_{i-1,i}$, we directly compute the variance of the estimated value $\hat{\rho}_{i-1,i}$:

$$\text{Var}[\hat{\rho}_{i-1,i}] = E[\hat{\rho}_{i-1,i}^2] - E^2[\hat{\rho}_{i-1,i}]. \quad (14)$$

The calculations for $E[\hat{\rho}_{i-1,i}]$ and $E[\hat{\rho}_{i-1,i}^2]$ are as follows:

$$\begin{aligned} E[\hat{\rho}_{i-1,i}] &= \int_{-b}^{\rho_{i-1,i}-b} \frac{x}{2be^{\epsilon_2/(t-1)} + 1} dx \\ &\quad + \int_{\rho_{i-1,i}-b}^{\rho_{i-1,i}+b} \frac{xe^{\epsilon_2/(t-1)}}{2be^{\epsilon_2/(t-1)} + 1} dx \\ &\quad + \int_{\rho_{i-1,i}+b}^{1+b} \frac{x}{2be^{\epsilon_2/(t-1)} + 1} dx - \rho_{i-1,i} \\ &= \frac{1+2b}{2(2be^{\epsilon_2/(t-1)} + 1)} + \frac{2bx(e^{\epsilon_2/(t-1)} - 1)}{2be^{\epsilon_2/(t-1)} + 1}. \end{aligned} \quad (15)$$

$$\begin{aligned} E[\hat{\rho}_{i-1,i}^2] &= \int_{-b}^{\rho_{i-1,i}-b} \frac{x^2}{2be^{\epsilon_2/(t-1)} + 1} dx \\ &\quad + \int_{\rho_{i-1,i}-b}^{\rho_{i-1,i}+b} \frac{x^2 e^{\epsilon_2/(t-1)}}{2be^{\epsilon_2/(t-1)} + 1} dx \\ &\quad + \int_{\rho_{i-1,i}+b}^{1+b} \frac{x^2}{2be^{\epsilon_2/(t-1)} + 1} dx \\ &= \frac{b^2}{3} + \frac{(2b+1)(b+1-3\rho_{i-1,i}^2)}{3(2be^{\epsilon_2/(t-1)} + 1)}. \end{aligned} \quad (16)$$

Based on (15) and (16), the variance term $\text{Var}[\hat{\rho}_{i-1,i}]$ can be derived as follows:

$$\begin{aligned} \text{Var}[\hat{\rho}_{i-1,i}] &= \frac{b^2}{3} + \frac{(2b+1)(b+1-3\rho_{i-1,i}^2)}{3(2be^{\epsilon_2/(t-1)} + 1)} \\ &\quad - \sigma_{i-1,i}^2 - 2\sigma_{i-1,i}\rho_{i-1,i}, \end{aligned} \quad (17)$$

where $\sigma_{i-1,i} = E[\hat{\rho}_{i-1,i}] - \rho_{i-1,i}$.

Theorem 4: R-DGP satisfies ϵ -LDP for each user.

Proof: R-DGP introduces a radius-constrained strategy based on DGP. Specifically, for a given trajectory T , Algorithm 3 divides the overall privacy budget ϵ into three components. The budget ϵ_3 is allocated to perturb the maximum distance d_{max} using SW mechanism, which satisfies ϵ_3 -LDP. Subsequently, the algorithm constructs a circular constraint region to strictly bound the spatial perturbation of the current point p_i . Crucially, the region D_i is centered at the previous perturbed point $\hat{p}_{i-1}$ with a radius equal to the perturbed maximum distance $\hat{d}_{max}$. Because this geometric construction relies exclusively on noisy outputs and never queries the raw data, it fundamentally qualifies as a post-processing operation. By the post-processing property of LDP, this procedure does not consume any additional privacy budget. The perturbations of the trajectory points and direction angles remain fully consistent with DGP, consuming privacy budgets of ϵ_1 and ϵ_2 , respectively.

Finally, according to the sequential composition theorem, R-DGP scheme satisfies ϵ -LDP, where $\epsilon = \sum_{i=1}^t \epsilon_1/t + \sum_{i=1}^{t-1} \epsilon_2/(t-1) + \epsilon_3$. $\square$

Since R-DGP inherits the perturbation strategy of DGP for trajectory points and directional angles, their error analysis processes are consistent with those of DGP, which are thus emitted here.

VII. EXPERIMENTAL EVALUATION

In this section, we will demonstrate the superiority of our proposed schemes through experiments. To effectively evaluate the effectiveness of proposed schemes, we conduct experiments on multiple metrics and various datasets.

A. Experimental Setup

Datasets: We conduct experiments on four benchmark trajectory datasets, namely NYC, CHI, CLE and CPS. These datasets are widely used in previous studies on trajectory privacy protection, which ensures the comparability and reproducibility of our results.

- **NYC:**¹ The NYC dataset consists of check-in trajectory data extracted from the Foursquare dataset in New York City [37], containing 101,044 records and 1,082 trajectories.
- **CHI:**² The CHI dataset consists of check-in trajectory data extracted from the Gowalla dataset [38] in Chicago, containing 36,094 records and 3,990 trajectories.
- **CLE:** The CLE dataset includes check-in trajectory data extracted from the Gowalla dataset [38] in Cleveland, containing 23,542 records and 1,463 trajectories.
- **CPS:** For CPS dataset, we follow the previous studies [10] and [11]. This dataset is based on the campus of the British Columbia University, containing 22,098 records and 4,000 trajectories.

We define the set of most popular POIs within the spatial domain $\mathcal{P}$. Specifically, for NYC dataset, we consider $|\mathcal{P}| =$

¹<https://www.nyc.gov/site/tlc/about/tlc-trip-recorddata.page>

²<https://www.kaggle.com/datasets/marquis03/gowalla/data>

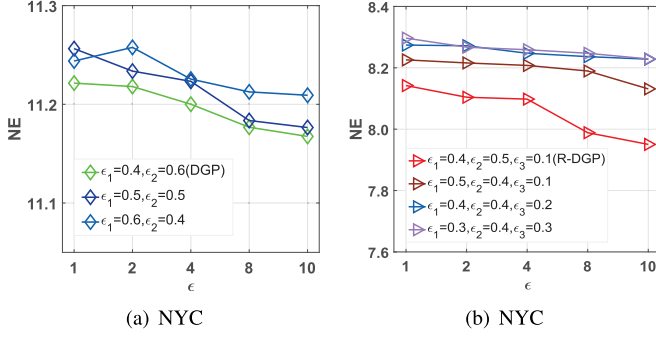


Fig. 5. Mean normalized errors (NEs) of different privacy budget allocation strategies.

2000 POIs, while for CHI and CLE datasets, the number of POIs is limited to $|\mathcal{P}| = 1000$. For CPS dataset, we take $|\mathcal{P}| = 262$ POIs. To ensure data authenticity and integrity, we do not split each trajectory but to retain original length. This strategy can maintain the natural movement patterns of each user's trajectory, thereby more accurately reflecting mobile behavior.

Baseline. In this paper, we propose a fundamental perturbation method called DGP which leverages the continuous directional information of trajectory points. Building on this, we also introduce an enhanced R-DGP scheme, which restricts the region of each trajectory point by radius constraint to improve data utility. To validate the effectiveness of our proposed schemes, we conduct a comparative analysis with existing state-of-the-art ATP mechanism and direct perturbation method k -RR.

- **ATP:** ATP mechanism is currently the state-of-the-art trajectory data collection method under user-level LDP. The method restricts the spatial regions of entire trajectory by duplicating each trajectory into two copies and alternately selecting pivot points to guide perturbation. Specifically, ATP uses the discretized direction information of pivot points to constrain the perturbation region of each trajectory point.
- **k -RR:** In our experiment, we use k -RR to represent the strawman method, which is one of the most basic location data perturbation mechanism under LDP.

TraCS [39] is a trajectory data collection method in continuous space. However, TraCS focuses on event-level privacy, while we focus on user-level privacy. Due to the different levels of privacy protection targeted, our schemes are not compared with TraCS method in the experiments.

Experiment Settings: We empirically determine a generalized privacy budget allocation strategy based on the utility impact of each perturbation step. Therefore, we evaluate multiple candidate allocation ratios on a large-scale NYC dataset, as shown in Fig. 5(a) and (b). The NYC dataset is chosen as the benchmark because it not only has a large number of records and the longest average trajectory length, but also has a very dense urban structure. The experimental results indicate that we do not need to select the optimal parameters for a specific dataset. Instead, we adopt a fixed allocation strategy, which can achieve stable and consistent performance.

Both the radius and direction angles fundamentally influence the spatial characteristics and overall utility of the trajectory. To

optimize data utility, we adopt a non-uniform privacy budget allocation strategy. Due to the cascading amplification effect caused by the angle errors, the overall trajectory shape is severely distorted. Therefore, we explicitly allocate more privacy budget to the angle perturbation part. Specifically, in R-DGP, we allocate $\epsilon_1 = 0.4\epsilon$ to perturb trajectory points and $\epsilon_2 = 0.5\epsilon$ to perturb direction angles. The remaining privacy budget, $\epsilon_3 = 0.1\epsilon$, is utilized to perturb the maximum distance. In DGP, the privacy budget ϵ follows a similar allocation principle: $\epsilon_1 = 0.4\epsilon$ is used to perturb trajectory points, and $\epsilon_2 = 0.6\epsilon$ for direction angles.

Environment: All approaches are implemented using Python 3.12 and NumPy 1.24 libraries. The experiments are conducted on a computer equipped with 2.10 GHz Intel Core i7 CPU and 32 GB RAM.

B. Utility Metrics

To more effectively validate the effectiveness of our schemes, we adopt four metrics from [10] and [6].

- **Mean Normalized Error (NE):** NE quantifies the average deviation between all original and perturbed trajectories, providing a quantitative measure for evaluating perturbation accuracy. A lower NE value indicates higher accuracy in trajectory preservation. The formula for NE is:

$$NE = \frac{1}{\mathcal{T}} \sum_{i=1}^{\mathcal{T}} \left(\frac{1}{t} \sum_{i=1}^t dist(p_i, \hat{p}_i) \right), \quad (18)$$

where $\mathcal{T}$ is the set of all trajectories, $dist(p_i, \hat{p}_i)$ denotes the Haversine distance between real point p_i and its corresponding perturbed point $\hat{p}_i$.

- **Preservation Range Queries (PRQ):** PRQ is used to measure whether the distance between each point in the perturbed trajectory and the corresponding true point is within a specified range. For example, a location PRQ might examine whether $\hat{p}_i$ is within 0.5km of p_j . For each point in trajectory T , we verify whether the perturbed point falls within threshold distance δ (km) of the corresponding true point:

$$PRQ = \frac{1}{\mathcal{T}} \sum_{i=1}^{\mathcal{T}} \left(\frac{1}{t} \sum_{i=1}^t dist(p_i, \hat{p}_i, \delta) \right) \times 100\%, \quad (19)$$

if $dist(p_j, \hat{p}_i) \leq \delta$, $dist(p_i, \hat{p}_i, \delta) = 1$; otherwise, $dist(p_i, \hat{p}_i, \delta) = 0$. A higher PRQ value indicates higher performance.

- **Average Count Difference (ACD):** ACD evaluates the accuracy of hotspots preservation by quantifying the absolute difference in access counts before and after perturbation. We consider a hotspot set H . For each hotspot $h \in H$, we record both actual visitor counts c and perturbed visitor counts $\hat{c}$.

$$ACD = \frac{1}{|H|} \sum_{h \in H} |c - \hat{c}|. \quad (20)$$

A lower ACD value indicates better preservation of hotspots structures.

- **Kendall's Tau Coefficient:** The Kendall's Tau Coefficient [6] evaluates the ranking differences in the popularity of locations. We select top n locations and compute the

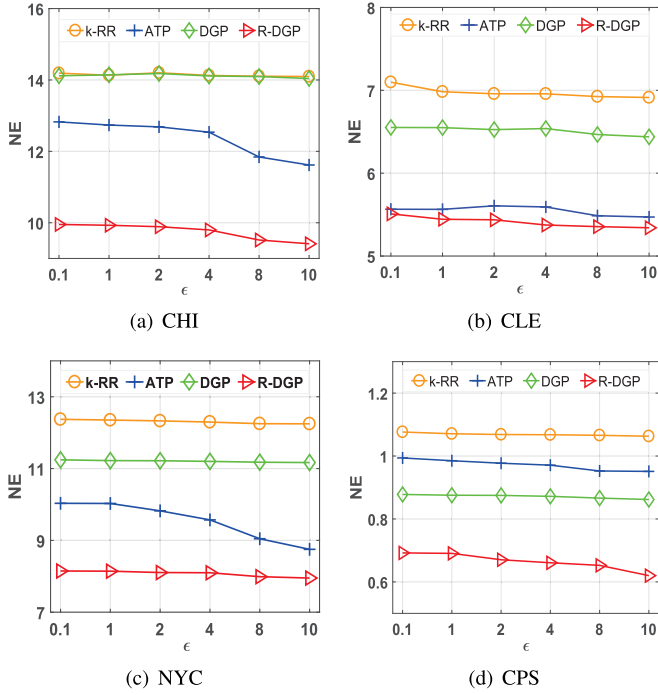


Fig. 6. Mean normalized errors (NEs) comparison under varying privacy budgets.

visit frequency of each location (in both real dataset $\mathcal{T}$ and perturbed dataset $\hat{\mathcal{T}}$). Then, we iterate through all pairs of locations. If the ranking order remains unchanged, the pair is classified as concordant. If the ranking order is reversed, the pair is classified as inconsistent.

$$KT = \frac{N_c - N_d}{n(n-1)/2}, \quad (21)$$

where N_c and N_d represent the number of concordant pairs and the number of discordant pairs respectively.

C. Experimental Results

We compared the performance of our proposed schemes and the benchmark method by adjusting different parameters under four metrics of NE, PRQ, ACD and KT.

1) *Mean Normalized Errors*: We conduct experiments on three real-world datasets (CHI, CLE and NYC) and one synthetic dataset (CPS) to evaluate the performance of our proposed schemes and baseline methods, as shown in Fig. 6. As the privacy budget ϵ gradually increases from 0.1 to 10, the amount of noise decreased and the NE value gradually decreases. Notably, benefiting from the aforementioned fixed budget allocation strategy, the results demonstrate that our schemes (represented by the red and green lines) outperform the baseline methods across all four datasets, demonstrating stronger stability.

Among the evaluation methods, we use k -RR to perturb trajectory points within entire domain. Due to its large perturbation space, it introduces a significant amount of noise, resulting in the lowest utility among all methods. The worst performance is observed on NYC dataset which contains 2000 trajectory points, as shown in Fig. 6(c). The ATP mechanism improves utility

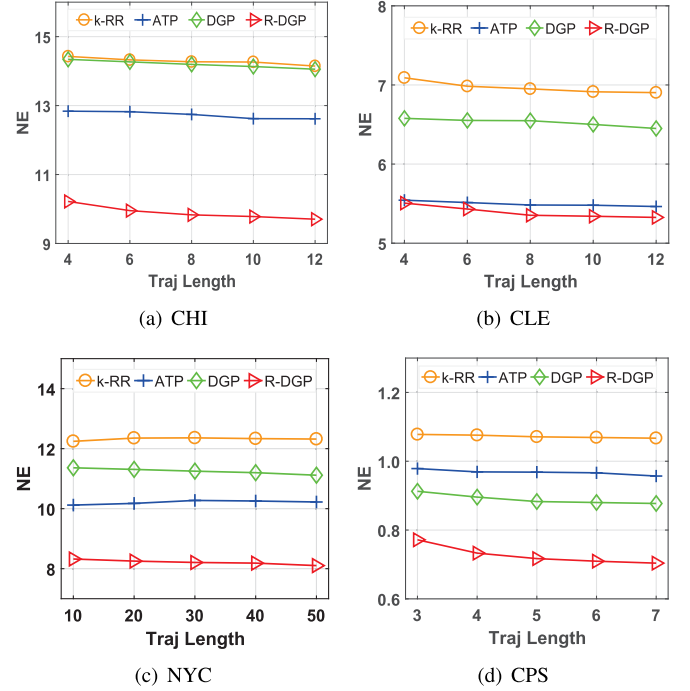
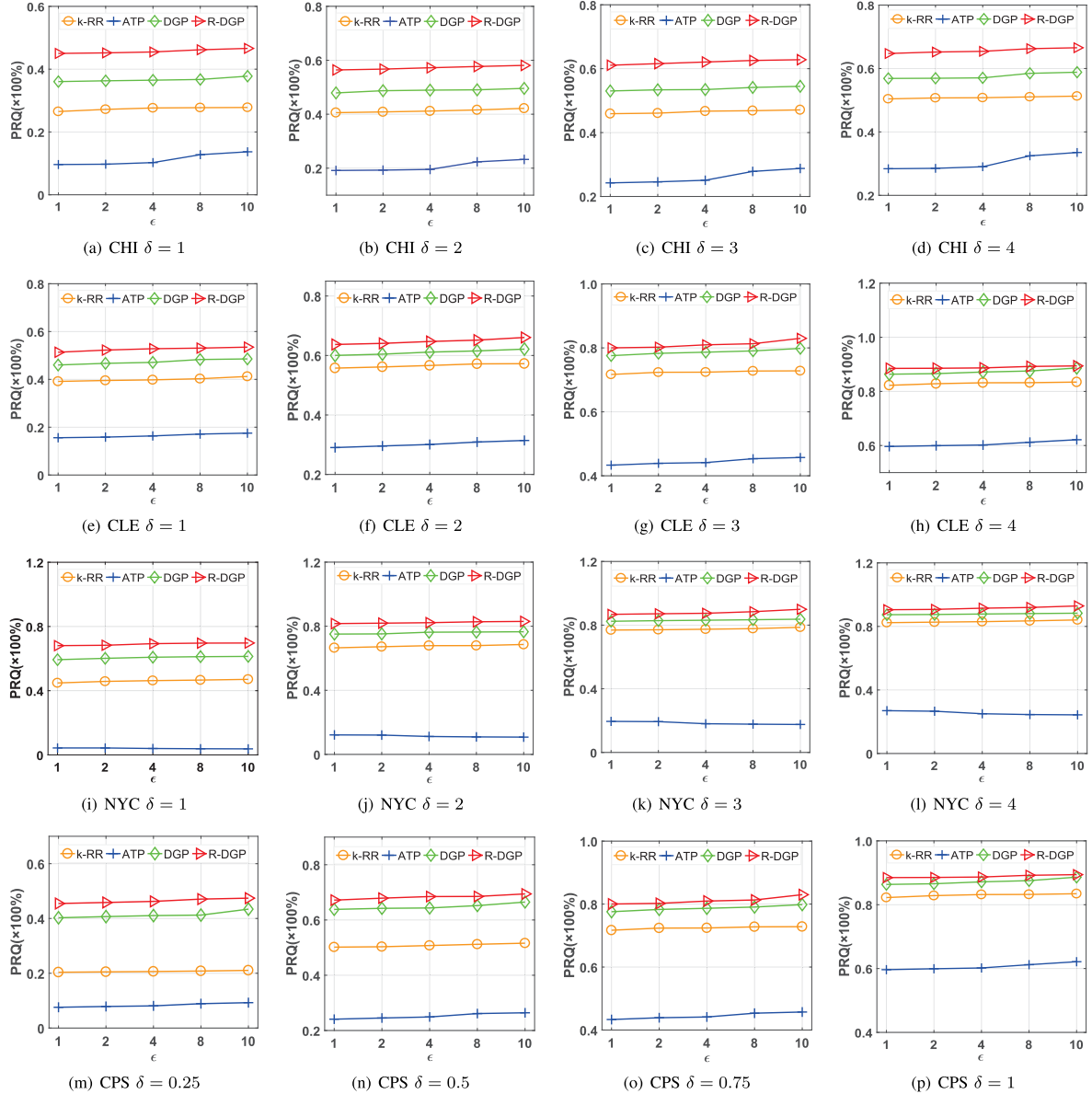


Fig. 7. Mean normalized errors (NEs) comparison under varying trajectory lengths.

by leveraging bidirectional direction information to determine the perturbation region of trajectory points. However, it may produce significant deviations due to perturbations in discrete directions. Additionally, since ATP allocates privacy budget to multiple tasks (duplicating trajectory, alternately sampling pivot points and discretizing the direction of pivot points), when ϵ is very small (e.g., $\epsilon = 0.1$ or 1), the noise introduced in each task becomes excessively large. In contrast, DGP determines the perturbation region of each trajectory point based on the direction of previous perturbed point, thereby enhancing utility. R-DGP further optimizes this approach by imposing stricter radius and direction constraint, effectively restricting the perturbation region. Therefore, based on the experimental results, R-DGP consistently outperforms DGP because longer trajectory lengths can typically make more effective use of radius and direction constraints.

2) *Impact of Trajectory Space Size and Trajectory Length*: We conduct extensive experiments on four datasets to evaluate the performance of our proposed schemes under different parameter settings. To analyze the impact of trajectory space size, we vary the number of trajectory space $\mathcal{P}$ while fixing privacy budget at $\epsilon = 4$ and observe the change in NE values. Due to the limited number of points on CPS dataset, this experiment is excluded. The number of $\mathcal{P}$ is gradually increased from 20% to 80% of total dataset, and we compare the performance of DGP and R-DGP, as shown in Table II. The experimental results demonstrate that the value of NE gradually increases as the number of trajectory space $\mathcal{P}$ increases, but the overall impact on performance remains relatively weak.

Following the experimental designs in [10] and [11], we experiment the impact of trajectory length on error results, as

Fig. 8. Range query queries (PRQs) under varying privacy budgets and δ values.TABLE II
COMPARISON OF NE WITH VARYING TRAJECTORY SPACE SIZE

$ \mathcal{P} $ (%)	CHI		CLE		NYC	
	DGP	R-DGP	DGP	R-DGP	DGP	R-DGP
20	4.784	3.792	2.248	1.962	8.526	6.647
40	4.795	3.833	2.256	1.972	8.536	6.677
60	4.822	3.853	2.262	1.987	8.569	6.735
80	4.857	3.888	2.284	2.031	8.669	6.777

shown in Fig. 7. Because the trajectory length distributions of NYC, CHI, CLE and CPS datasets are [3, 1722], [2, 1204], [2, 1122] and [3, 8], respectively, we set different length ranges for each dataset. Throughout these experiments, the privacy budget is fixed at $\epsilon = 4$. The experimental results indicate that the overall error of R-DGP decreases as trajectory length increases.

This trend is likely due to the enhanced impact of directional constraint on longer trajectories. Furthermore, R-DGP consistently outperforms ATP mechanism on all datasets, exhibiting higher stability. These findings further validate the effectiveness of R-DGP under different trajectory lengths.

3) *Preservation Range Queries*: In the PRQ experiment, we set the parameter δ to 1, 2, 3 and 4 for CHI, CLE and NYC datasets. For CPS dataset with a denser distribution of trajectory points, we set δ to 0.25, 0.5, 0.75 and 1. This experiment aims to evaluate the change of PRQ values as privacy budget ϵ increases from 1 to 10. The PRQ values of four datasets exhibit different characteristics due to differences in dataset size, trajectory movement range and user's trajectory length. However, we observe a consistent trend where the overall accuracy distribution remains similar and the PRQ values show an increasing trend when δ increases from 1 to 4 (or from 0.25 to 1 on CPS). This

TABLE III
COMPARISON OF ACD FOR DIFFERENT METHODS

$ H $ (%)	CHI			CLE			NYC			CPS		
	ATP	DGP	R-DGP	ATP	DGP	R-DGP	ATP	DGP	R-DGP	ATP	DGP	R-DGP
20	32.929	42.629	40.075	16.985	19.587	18.497	40.176	22.306	21.377	51.973	32.488	27.684
40	25.804	26.393	25.913	15.042	14.008	13.462	41.622	25.545	24.926	49.728	31.784	36.819
60	24.303	22.257	22.131	14.895	12.713	12.411	39.646	28.235	27.361	48.025	30.987	26.329
80	22.608	21.881	21.135	14.295	13.621	13.237	36.567	30.441	28.433	47.459	31.337	25.898

TABLE IV
COMPARISON OF KT COEFFICIENT FOR DIFFERENT METHODS

ϵ	CHI			CLE			NYC			CPS		
	ATP	DGP	R-DGP	ATP	DGP	R-DGP	ATP	DGP	R-DGP	ATP	DGP	R-DGP
1	0.1564	0.1861	0.2024	0.0529	0.0985	0.1306	0.0640	0.0041	0.0217	-0.0922	-0.0432	-0.0238
2	0.1638	0.1937	0.2196	0.0877	0.1098	0.1457	0.0194	0.0086	0.0301	-0.0857	-0.0365	-0.0204
3	0.1889	0.2068	0.2307	0.0484	0.1151	0.1571	0.0975	0.0132	0.0476	-0.0742	-0.0320	-0.0138
4	0.2041	0.2181	0.2482	0.0759	0.1195	0.1846	0.1006	0.0142	0.0615	-0.0726	-0.0106	-0.0106

phenomenon can be attributed to the gradually decreasing amount of introduced noise. It is evident from Fig. 8(a)–(p) that R-DGP consistently outperforms all other methods in terms of range query protection.

4) *Evaluation on Practical Applications:* To validate the practical utility of our schemes, we conduct hotspot preservation experiments on four datasets. The ACD provides a measurement to assess the extent to which noisy datasets retain the distribution and intensity of hotspots. With the privacy budget fixed at $\epsilon = 4$, we vary the size of the hotspot set $|H|$ from 20% to 80% of total locations. As demonstrated in Table III, both DGP and R-DGP consistently outperform the baseline ATP method, with their advantages becoming particularly pronounced in scenarios featuring denser hotspot distributions and larger-scale datasets. Notably, R-DGP exhibits strictly superior performance over DGP. This indicates that enforcing more precise spatial constraints on the trajectory perturbation regions effectively enhances the fidelity of hotspot preservation.

We utilize the Kendall's Tau (KT) coefficient to evaluate the consistency of popularity rankings. Specifically, we evaluate the top- n most popular locations, setting $n = 100$ for CHI and CLE datasets, $n = 200$ for NYC dataset, and $n = 50$ for CPS dataset. The metric quantifies the ranking difference in top- n most popular positions between the original and perturbed dataset. A KT value close to 1 indicates that the privacy mechanism has a minimal impact on data utility. Conversely, if the value is close to 0 indicates that the perturbed rankings are nearly random, reflecting significant utility loss due to injected noise. As shown in Table IV, our schemes preserve data utility much more effectively than existing methods, achieving a better balance between privacy protection and utility.

VIII. CONCLUSION

In this paper, we investigate the trajectory collection problem under the LDP framework and propose a direction-guided perturbation scheme called DGP. DGP utilizes the movement path of trajectory and the direction angle of adjacent points to determine the perturbation range, thereby reducing noise.

Furthermore, we introduce a radius constraint and extend it to R-DGP to ensure the spatial consistency of the perturbed trajectory. Extensive experiments demonstrate the superiority of our proposed schemes in balancing privacy protection and data utility. As our future work, we aim to optimize our schemes and expand its application to more complex trajectory scenarios to enhance its adaptability in real-world environments.

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